ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 3

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Paper 1 style (theory). Section A: objective questions (30) · Section B: atomic theory, SI units and workshop safety, Ohm's law, power, energy cost and series-parallel networks, resistor colour code, capacitors, semiconductors and LEDs, number systems, logic gates and signals (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The SI unit of electric charge is the …

A volt. B coulomb. C ampere. D joule.

1.1.2 The particle with a negative charge is the …

A proton. B neutron. C electron. D nucleus.

1.1.3 Ohm's law is written as …

A V = I ÷ R B V = I × R C I = V × R D R = V × I

1.1.4 The unit of electrical power is the …

A watt. B ohm. C farad. D henry.

1.1.5 A resistor with the colour bands brown, black and red has a value of …

A 100 Ω. B 10 kΩ. C 102 Ω. D 1 kΩ.

1.1.6 The best conductor of the following is …

A rubber. B glass. C copper. D plastic.

1.1.7 The binary number 1010 is equal to the decimal number …

A 8 B 10 C 12 D 5

1.1.8 The longer leg of a new LED is the …

A cathode. B gate. C anode. D base.

1.1.9 The fire extinguisher that may be used on an electrical fire is …

A carbon dioxide (CO_2). B water.
C foam. D a wet cloth.

1.1.10 The gate whose output is 1 only when both inputs are 1 is the …

A OR gate. B NOT gate. C NOR gate. D AND gate.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Electric current is measured in amperes.

1.2.2 Voltage is the flow of electrons through a conductor.

1.2.3 Resistors connected in series all carry the same current.

1.2.4 The total resistance of resistors in parallel is larger than the largest resistor.

1.2.5 A fuse protects a circuit by melting when the current is too high.

1.2.6 Silicon is a semiconductor.

1.2.7 A capacitor lets direct current flow through it continuously.

1.2.8 The hexadecimal digit F is equal to 15 in decimal.

1.2.9 A NOT gate has two inputs.

1.2.10 Safety goggles should be worn when soldering.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Unit of resistance	A	capacitor
1.3.2	Rate at which electrical energy is used	B	truth table
1.3.3	Electrical "push" that makes electrons flow	C	ohm
1.3.4	Material that allows almost no current to flow	D	soldering iron
1.3.5	Component that lets current flow in one direction only	E	insulator
1.3.6	Component that stores electric charge	F	voltage
1.3.7	Number system with base 2	G	binary
1.3.8	Shows the output of a gate for every combination of inputs	H	diode
1.3.9	Tool used to melt solder onto a joint	I	multimeter
1.3.10	Instrument that measures voltage, current and resistance	J	power

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: ATOMIC THEORY, UNITS AND SAFETY

2.1 Describe the structure of an atom. Name the THREE particles and state the charge and position of EACH. (3)

2.2 Explain why metals such as copper are good conductors of electricity. (2)

2.3 Convert the following: (a) 4 700 Ω to kΩ; (b) 0,025 A to mA; (c) 2,2 µF to F; (d) 3 kV to V. (4)

2.4 Give FIVE safety rules for working in an electronics workshop. (5)

[14]

QUESTION 3: OHM'S LAW, POWER AND NETWORKS

3.1 Three resistors of 100 Ω, 220 Ω and 680 Ω are connected in series across a 12 V supply. Calculate:

3.1.1 The total resistance (1)

3.1.2 The current in the circuit (2)

3.1.3 The voltage across the 680 Ω resistor (2)

3.2 A 6 Ω and a 3 Ω resistor are connected in parallel across a 12 V supply. Calculate the total resistance, the total current and the current through the 6 Ω resistor. (4)

3.3 A 4 Ω resistor is connected in series with a parallel combination of 6 Ω and 12 Ω, across a 24 V supply. Calculate:

3.3.1 The total resistance (2)

3.3.2 The total current (1)

3.3.3 The voltage across the parallel combination (1)

3.3.4 The current through the 12 Ω resistor (1)

3.4 A 60 W soldering station is used for 5 hours a day for 20 days. Electricity costs R2,50 per kWh.

3.4.1 Calculate the energy used in kWh and the cost. (3)

3.4.2 Use Joule's law to calculate the heat produced in a 10 Ω resistor that carries 2 A for 60 seconds. (3)

[20]

QUESTION 4: COMPONENTS AND SEMICONDUCTORS

4.1 Determine the value and tolerance of a resistor with the colour bands yellow, violet, orange and gold. (3)

4.2 Two 10 µF capacitors are connected (a) in parallel and (b) in series. Calculate the total capacitance in EACH case. (3)

4.3 Explain what doping is, and how N-type and P-type semiconductor material are made. (4)

4.4 Draw the circuit symbols for: (a) a diode; (b) an LED; (c) an electrolytic capacitor; (d) a single-pole switch. (4)

4.5 Explain why an LED needs a series resistor, and calculate the resistor needed for a 2 V, 20 mA LED on a 9 V battery. (4)

[18]

QUESTION 5: NUMBER SYSTEMS, LOGIC AND SIGNALS

5.1 Convert: (a) the decimal number 45 to binary; (b) the binary number 11010110 to decimal; (c) the decimal number 214 to hexadecimal. Show your working. (6)

5.2 Draw the truth tables for a 2-input OR gate and a 2-input NAND gate. (4)

5.3 A buzzer (Y) sounds when push button A AND push button B are both pressed, OR when the emergency switch C is closed.

5.3.1 Write the Boolean expression for Y. (2)

5.3.2 How many rows will the truth table have? (1)

5.3.3 What is the output when A = 1, B = 0 and C = 1? (1)

5.4 Explain the difference between an analogue signal and a digital signal, and give ONE example of each. (4)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER